Multi-agent systems for energy optimization in smart industries

Case study: Milford Haven Port

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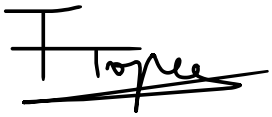
Signed: 

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Abstract

With rising concerns about energy use, companies and industries around the world are seeking solutions to decrease energy-related expenses and carbon footprints. The rising cost of energy has made energy conservation and optimization a significant priority for organizations that have begun to employ more intelligent energy management approaches to reduce costs. There is a strong need to assess energy usage in fisheries ports to find areas for improvement and optimize energy consumption while reducing carbon emissions. This project will investigate how an agent-based coordination framework for managing processes and resources can be implemented for smart industries. The project will involve the development of an agent system to model appliances and buildings, as well as the establishment of smart energy clusters.

The structure of this project starts with the abstract which describes the main goals of the report. It specifies the general goal of using modeling, simulation, and smart grid technologies to improve energy usage and minimize costs in the fish sector while helping seaport decarbonization. The next section is the introduction which examines the issues facing the fishing industry, particularly the continued growth in energy costs and pressure to address climate change. It also offers perspective on the importance of the fishing sector and the need to optimize energy consumption and minimize emissions. Section 2 is the literature review that seeks to identify important difficulties, examine prior research, and provide a full overview of the current state of research. The next section is the methodology which describes the development and implementation of MATLAB simulation software to optimize energy usage and scheduling under multiple situations, including individual building optimization, cluster optimization, and peer-to-peer energy optimization methodologies. Section 4 is the evaluation which explains each of the scenarios and their pseudo-codes. The results section is next, which analyzes the outcomes of the simulation for each scenario. Finally, the conclusion describes the significance of the work, the rigor, and its applications.

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1. Introduction

The fishing industry is one of the world's most traded industries, with rising demand and supply for fisheries and aquaculture products. However, the fishing sector confronts considerable challenges, including rising energy costs. This comes at a time of increasing international pressure to address climate change and reduce carbon emissions in various businesses. This paper intends to make use of modeling, simulation, and smart grid technologies to optimize energy usage and minimize costs in the fishing industry, while also contributing to the decarbonization of seaports.

1.1 Climate change and energy consumption

In recent years, there has been a big change in the earth's climate as a result of global warming and an increase in carbon dioxide levels. As illustrated in Figure 1, one of the fundamental causes of climate change is the excessive use of primary energy sources because of the Industrial Revolution, which resulted in a significant amount of carbon being emitted into the atmosphere. Using resources such as fossil fuels has resulted in a rise in greenhouse gas emissions into the atmosphere, which is expected to contribute to global warming, climate change, and the ozone layer. Carbon emissions are mostly caused by the combustion of fossil fuels.

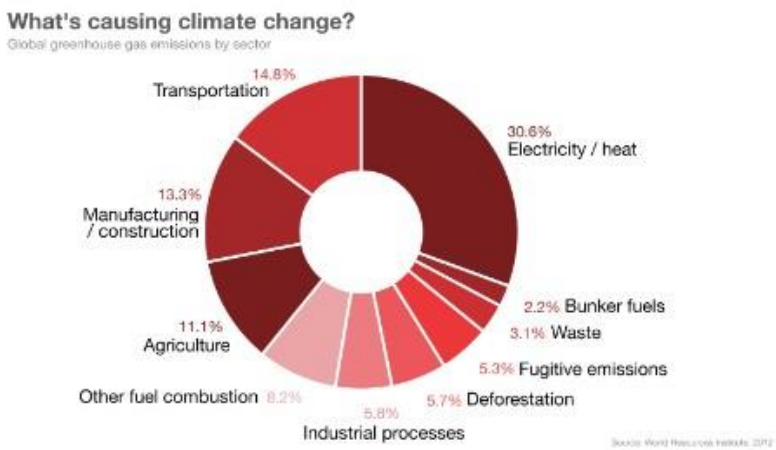

Figure 1: Role of energy consumption on climate change (World Resources Institute,2012)

Currently, they are utilized to generate energy in automobiles, generators, trains, and airplanes. The fuel combustion process in engines creates exhaust gases that contain air pollutants such as sulfur dioxide (SO2), particulate matter (soot), nitrogen oxide (NOx), and a considerable amount of carbon dioxide (CO2) [48]. Climate change is fundamentally linked to energy use. As a result, the necessity to use energy for various reasons in different sectors around the world must be considered. The global demand for primary energy sources has risen due to an increase in population. Meanwhile, the growing number of power users and the Industrial Revolution in developing countries have contributed greatly to global economic growth during the last Although. It is predicted that the world's energy demand will rise during the next decade due to population growth and increased industrial activity. The industrial sector is anticipated to raise its energy

usage by 40% by 2040. With new products and increasing resources, the composition of production is shifting, resulting in slower growth in industrial energy and material use than in the overall economy. Technological innovation tends to minimize the total energy and material needs for each industrial operation. This comparatively optimistic scenario meets significant challenges; industrial decision-makers heavily discount future energy expenses for a variety of reasons. Furthermore, many slow-growing heavy industries may lack the finances and technological capacity to invest in the most effective new manufacturing processes. For these reasons, it is critical to adopt public policies that support the development and diffusion of more efficient process technologies, as well as to aid energy-intensive sectors to modernize, such as through utility demand-side management programs. It is critical to increase energy efficiency across all industrial sectors. The ultimate goal is to reduce greenhouse gas emissions, ensure energy security, and boost industrial output and competitiveness. In the final stage of the manufacturing chain, dedicated customers in industrialized countries have become aware of the issue of global warming, modified their consumption habits, and begun to demand green and sustainable products and services. The uncertainty surrounding global fossil energy reserves and their long-term availability raises awareness of the need for the development of clean and renewable energy resources, as well as improved energy use to avoid waste [38].

1.2 Energy consumption in fish industries

This study is based on the topic of smart industries, specifically the coordination of processes and resources utilizing an agent-based framework. The fishing industry is one of the most energy-intensive methods of food production, and there is an increasing demand for intelligent systems that can efficiently manage and optimize industrial processes. Fish processing facilities are highly complicated, with major environmental impacts such as air quality and carbon emissions. They have the highest running expenses and energy usage due to constant refrigeration, air conditioning, and ice-making equipment. The fishing industry's high energy consumption and usage of fossil fuels have led to an increase in carbon emissions over time. Using renewable energy reduces carbon emissions and reduces reliance on fossil fuels, offering long-term benefits. Intermittent renewable energy supplies can negatively affect fish industry operations. The continuous loss of wild fish supplies is putting a further burden on the aquaculture industry in terms of supplying fish and seafood to global markets. Despite the fact that aquaculture is more diverse than other agricultural sectors, there are significant pressures on the industry to continue innovating in order to enable sustainability, such as increased fish production, improved species selection, disease mitigation, waste reduction, environmental pollution prevention, and global employment opportunities [39].

This report aims to evaluate the simulation's efficiency, adaptability, and resource optimization while gathering insights and feedback to help improve the coordination framework. This simulation-based technique is used to explore the ability of renewable energy sources, particularly local solar farms, to supply local power demand. Energy usage in fish processing businesses can be divided into two modes: (i) direct use, such as lighting, heating, and box washing machines, and (ii) indirect use, which involves converting power to another kind of energy, such as cooling, freezing, and equipment. Energy consumption varies depending on factors such as the age of the

production plant, the level of automation, and the maturity of production processes. The other parameters that can influence the energy usage for fish industries are (i) seasons when the amount of fishing can increase the energy demand; (ii) weather, which can have a significant effect on the total processed fish during a year; and (iii) the number of fisheries and boats, which can also affect the total energy use. This project suggests that new smart energy systems can address the needs of fish processing industries and ports by utilizing renewable energy and smart energy management. Seaports are under pressure to decrease their carbon footprint, improve energy efficiency, and maintain global competitiveness. Continuous monitoring of energy consumption in seaports is crucial for managing rising energy costs and fuel demand. Fishery buildings at ports require energy-intensive equipment, including refrigerators, air conditioning, and ice makers. This leads to higher energy costs and indirectly increases carbon emissions. To improve sustainability and competitiveness in the energy market, most fishery buildings must be modernized. In 2011, fishing buildings used around 40 billion liters of fuel and produced 179 million tons of carbon emissions. The lack of updated energy efficiency regulations leads to decreased energy usage and lower emissions. Fishery buildings provide around 4% of worldwide greenhouse gas emissions. Human consumption of seafood has risen from 20 million tons in 1950 to over 136 million tons by 2014. The rising consumption of seafood has led to higher carbon emissions due to the industry's high energy usage. Finding efficient, sustainable, and economical energy sources is vital for reducing energy costs and limiting carbon emissions in fishing buildings [40]. The fishing sector's energy consumption may vary depending on activities at sea, breeding, food preparation, or processing. Various types of vessels are used in the fishing industry, including purse seiners, trawlers, longliners, gillnetters, crabbers, and others, depending on the activity and purpose. As a result, energy consumption varies accordingly. Fuel is a significant expense for most fishing operations across all production levels. Its importance is defined by the distance to the fishing grounds, fishing activity, and type, vessel, and management issues. For example, it is lower in aquaculture than in capture production due to the longer duration of time fishermen spend at sea. Given inflation, aquaculture must deal with greater maintenance and operational costs, thus finding economically viable and environmentally benign ways is a necessity. Capital expenses may be higher at times, but in the long run, given price fluctuations over the year, certain alternative techniques are preferable [41].

1.3 Seaports

Seaports contribute significantly to global economic growth. Seaports have historically played a significant role in facilitating import and export movements between countries, as well as shaping international trade and global supply networks. As a result of the congestion of commercial activity from ships and container lifting and unloading operations, as well as the nearby presence of seafood industries, modern seaports, and their neighboring cities have grown carbon-intensive and have significant pollution levels. As a result of this worldwide accord, some countries began to implement regulations, legislation, and policies to encourage the use of renewable energy sources (RES). Mitigation strategies include integrating energy efficiency and renewable energy, investing in energy savings, and transitioning to zero-energy buildings [43]. Energy efficiency is becoming more appealing to ports and terminals as they see that significant energy savings may be achieved through operational rationalization, the deployment of new technology, and the use of renewable

energy sources. This is part of many ports' increased focus on improving their green image and developing environmental strategies to better manage negative externalities and stakeholder relations, whether for a competitive advantage, to improve their strategic management processes, or in response to shipping industry pressure. The problem of energy efficiency is not new, but in the case of ports, despite the growing number of initiatives focused on energy savings and renewable energy development, no port-specific study has addressed this issue. Although energy efficiency literature has emerged in a variety of industries, including manufacturing, small and medium-sized companies (SMEs), universities, and shipping, research in this area is far from complete [44].

1.4 Smart micro-grids

Smart microgrids can alleviate this problem by storing or sharing extra energy within the local cluster. The European Technology Platform Smart Grids (ETPS) first introduced the notion of smart grids in 2005. Smart grid technology refers to the use of computer systems to transmit power from suppliers to consumers (via control automation, continuous monitoring, and distribution system optimization). The goal of a smart grid is to save energy, save costs, and boost reliability by using smart microgrid-controlled distribution networks that act independently. The advantage of smart grids is that they allow renewable energy sources to interact with local energy sources, increasing power dependability and quality throughout the system. Consumers play an important role in a smart grid because they may use energy systems to prevent energy waste and loss, reduce peak power use, and increase energy efficiency. Smart grid strategies can also help to cut operating costs. They integrate the behavior of all users, including suppliers and consumers, to provide sustainable power systems with low losses, high quality, supply security, and safety. Smart microgrids improve system stability and reduce interaction with the grid, resulting in fewer transmission losses. Integrating renewable energy resources with energy storage and load in a single network and establishing a smart energy management system helps control the intermittent nature of renewable energy sources (RES). There are major advantages to employing microgrids to govern Distributed Energy Resources (DERs). Some of them include lower overall electrical energy consumption and negative environmental impacts, a more reliable energy supply and higher energy efficiency than centralized grids, lower transmission losses, flexible voltage and frequency control, and high-power supply security. Another key issue is that microgrid power systems can provide backup power to the utility grid during natural catastrophes or urgent conditions. Solar photovoltaic energy is the fastest-growing renewable energy source in terms of electricity generation. The total installed solar PV capacity globally was 403 GW at the end of 2017, with approximately 90 GW of solar photovoltaic capacity installed in 2017. The rising use of solar photovoltaic energy sources raises concerns about reliability, power quality, frequency stability, and other aspects of the centralized grid system. As a solution to these difficulties, microgrid systems with battery storage units emerge as one of the utility grid's primary partners in sharing expanding solar PV power. This is a significant advantage of microgrids. Microgrid systems are particularly profitable for users because the electricity demand during peak hours may be derived utilizing the stored energy in the battery bank. Microgrids can be used to provide electrification for rural communities and places that the main grid cannot reach. This will have a favorable socioeconomic influence on rural communities. The usage of microgrids has many good

environmental consequences because it reduces the use of fossil fuels and increases the share of environmentally friendly energy sources such as solar, wind, ocean wave, geothermal, and so on. When it comes to emissions and water utilization (both during power generation and technology assembly), microgrids outperform utility grids that use traditional power-generating methods (coal, diesel) [42].

1.5 Project aims and objectives

The following research questions set the basis of work required for this project. There are three research questions explored:

- How does the application of MAS facilitate a shift in the way energy-intensive industries manage their energy usage?
- What are the advantages and disadvantages of implementing MAS in smart energy communities?
- How can agents effectively portray and account for a variety of elements such as seasonal variations, weather conditions, and consumer behaviors, in order to maximize energy efficiency?

The aims and objectives are based on the research questions and will be explored throughout the project:

Aim 1: Conduct research on the impact of energy use in energy-intensive industries

- To convert seaports and fishing ports into more efficient and cost-effective facilities by raising understanding of energy systems, automating processes, and leveraging available resources.

Aim 2: To investigate how MAS can be implemented to optimize energy efficiency

- Create an agent-based coordination framework for managing processes and resources in smart industries.
- To investigate the ability of MAS to model various acts to manage energy consumption while taking into account economic considerations, non-invasive approaches, and user comfort.

Aim 3: To explore how the use of MAS framework can reduce energy costs and carbon emissions

- Evaluate the MAS simulation's efficiency, adaptability, and resource optimization while gaining insights to improve the coordination framework and discover areas for energy savings, cost reduction, and emissions reduction.
- Evaluate how using MAS can benefit future work.

The research questions directly align with the objectives and findings presented in this project.

The following hypothesis is explored: “The use of multi-agent systems (MAS) and optimization approaches can considerably improve energy management, save costs, and help to decarbonize the fishing sector and seaports”.

2 Literature Review

The literature review intends to provide a comprehensive understanding of the present state of research in the field and identify gaps that the project may solve to optimize energy consumption and minimize costs in the fish industry while contributing to the decarbonization of seaports.

2.1 Multi-agent systems

A MAS is a system of agents that interact, communicate, and coordinate to achieve specific objectives. “An agent is a computer system that is situated in some environment, and that is capable of autonomous action in this environment to meet its design objectives”. Wooldridge's [7] idea defines an agent as a computer system that may behave autonomously in its surroundings to achieve its design objectives. The concept of an agent has been widely employed in social sciences since it is one of the alternatives to mathematical models for modeling various actions and simulating them. In the field of building energy optimization, this ability to simulate social behaviors is a very useful tool to know whether achieving the marked objective (reduced energy consumption) according to the set of rules (economic parameters cannot be jumped, non-invasive methods must be used, etc.) does not affect user comfort (lighting, heating, cooling, and ventilation within the values desired by the user).

However, the MAS has evolved so that it now serves more than just model situations. The agents' skills have improved as the concept and development of the MAS has progressed. Corchado and Laza [8] described one of the strategies being developed by the community of agent investigators to add learning capacity to belief-desire-intention (BDI) agent architectures. They used case-based reasoning (CBR) methods to help agents solve problems, allowing them to adapt to changes in the environment and suggest new potential answers. Different works emerged as a result of the agents' adoption of learning techniques, as presented by Di Mascio [5], where an agent-based approach was developed for children aged seven to eleven years old with hearing difficulties, to improve their reading ability, thus improving their school performance—thanks to the agents' ability to learn about each child's situation and provide different games and educational activities that promote reading. De la Prieta [6] describes an agent-based architecture that takes advantage of Cloud Computing platforms to address unresolved concerns in the Learning Object paradigm. The granting of learning capabilities to agents has allowed MAS to reason about different possibilities (different configurations of lighting, heating, cooling, or ventilation in an Intelligence Building, for example), autonomous decision-making, learning new situations, and errors from the past. These new characteristics and abilities have enabled MAS to be used in a wide range of applications, including drug identification systems for intoxicated drivers, obtaining genes with varying levels of expression indicating a disease or its application, and classifying faces based on gender and age.

2.2 Energy usage in seaports

The fishing business is one of the world's most traded products, with increasing demand and supply of fisheries and aquaculture products. According to the Food and Agriculture Organization (FAO), global fish production in 2016 was approximately 171 million tonnes, compared to 19.3 million tonnes in 1950 [34]. Global investment in fish products has expanded from € 8 billion to € 133 billion during the last 40 years [35]. Fish processing involves preparing fish and seafood for final delivery to customers. Fish processing is divided into two stages: primary processing (cutting, filleting, picking, peeling, washing, chilling, packing, heading, and gutting) and secondary processing (brine, smoking, cooking, freezing, canning, deboning, breading, vacuum, and controlled packaging, and ready meal production [36]. Energy consumption has a significant environmental impact on fish processing companies, as it is necessary for several on-site activities. Electricity is utilized in the fish industry to power processing equipment, lighting, cooling, and freezing. Energy consumption in the fish industry varies depending on factors such as plant age, automation, and manufacturing methods. Rising fuel and energy costs have significantly impacted the worldwide food processing industry in recent decades. The amount of fuel and energy required varies per process. Energy and fuel use directly impact fish production. The cost of energy and fuel is sensitive to the revenue generated by fish trading procedures. Energy costs might be considerable in the fishing industry. Fuel equivalent levels refer to the amount of fuel and power consumed directly in manufacturing, including raw material collection, processing, and distribution. The global energy cost for the fisheries industry is around € 9.85 billion. The anticipated energy usage of blast freezers ranges from 10.9 to 20.7 GWh, with a cost of 1.7 to 3.2 million at USD 0.15/kWh. Cold stores consume around 45 GWh, costing $6.8 million at USD 0.15 per kWh [46].

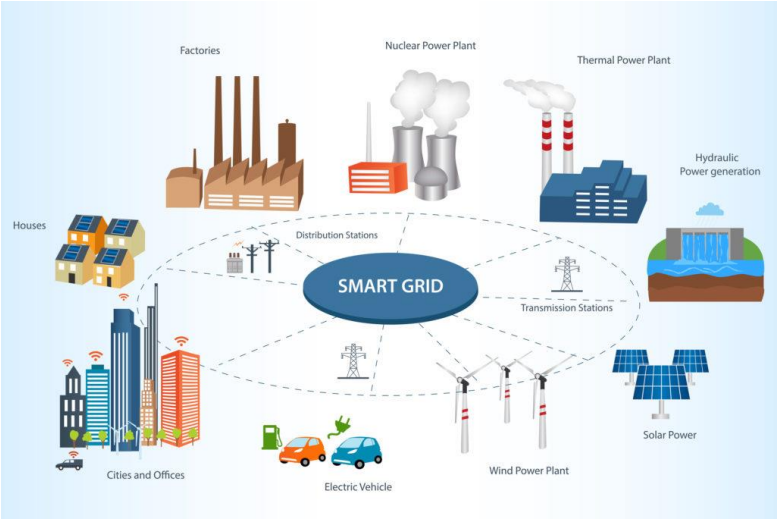

Figure 2: Optimizing smart energy infrastructure in smart ports, figure taken from IEEE.

The fish processing industry consumes a lot of energy and contributes significantly to carbon emissions. There are numerous methods for preparing fish. The process begins with catching fish and ends with delivering the product to end users. The fish industry consumes significant amounts

of electricity due to cooling and freezing processes. According to Ireland's Seafood Development Agency (BIM), refrigeration and air conditioning account for around 15% of global energy usage. Inefficient cooling systems in the fish industry can cause significant emissions of refrigerant gasses like ammonia. Cold storage in fish businesses accounts for approximately 80% of total energy usage, lasting 24 hours. Processing equipment might take more energy during warmer seasons, leading to higher energy consumption. The fish industry faces challenges such as maintaining cold storage and equipment, as well as rising energy costs that may not cover the cost of catching land fish. As a result, its impact will be felt across the system. Furthermore, global decarbonization efforts, particularly in the industrial sector, may need to align with local policies aimed at reducing carbon emissions, reducing energy consumption, and promoting renewable energy [47].

According to the search results, research on seaport energy applications is heading toward the deployment of smart technologies that will improve the operational efficiency of port facilities, as illustrated in Figure 2. The Royal Port of Stockholm [3] is developing smart house architecture to reduce carbon emissions and promote sustainability in the port district. The project aims to improve energy usage by allowing consumers to interact with utilities. A study by [1] created a smart microgrid for a fisheries port to use renewable energy and cut carbon emissions. Niglia [4] emphasized the importance of using smart grids to secure critical infrastructure in marine applications. This study emphasizes the necessity of security for smart grid technology at seaports. This study highlights the significance of securing smart grid technologies at seaports to prevent disruptive cyber-attacks. Molavi et al. [9] examined the advantages of implementing a microgrid at a port site. This study used a multi-stage stochastic mixed integer programming model to assess the effectiveness of implementing a microgrid at a port location using various parameters. The authors stated that implementing a microgrid can harm port activity. Arena et al. [10] explored the possibility of generating energy from sea waves and charging electric automobiles at seaports.

Ramos et al. [11] investigated using a tidal farm to power a port location. The analysis concluded that 25 tidal turbines can supply the port's electricity demands. Lam et al. [12] studied the impact of utilizing an EnMS (set of interrelated or interacting elements to establish an energy policy and energy objectives, and processes and procedures to achieve those objectives) at seaports. Implementing a smart EnMS at seaports can have economic and environmental benefits for both the port and neighboring communities, according to the authors. Parise et al. [13] suggest that implementing an energy master plan at a port can enhance the efficiency of its operating systems. Implementing a smart EnMS can optimize energy usage at seaports by utilizing smart grids, microgrids, and shore-to-ship power supplies. Acciaro et al. [14] found that implementing an energy management system (EnMS) in ports can reduce energy costs and improve energy efficiency. Sarabia-Jácome et al. [15] created a port data space to prevent interoperability of stakeholder information systems. The results indicate that the seaport data space enhances decision-making among seaport departments.

The 2020 European plan focuses on three key features: smart, sustainable, and inclusive growth [16]. One program aims to transform ports into smart ports. The smart port project aims to implement a new energy model and value proposition using innovative technologies that have little operational and environmental impact. Buiza et al. [17] defined a smart port as focusing on

operational and energy efficiency, competitiveness, and environmental impact. The smart port concept utilizes IoT to continuously collect information on seaport activities and AI and big data technology for decision-making. This strategy creates a durable data storage system that enables several entities to access important data based on an agreed-upon governance model, leading to improved operational efficiency. Seaports are a multidimensional system that combines economic functions, infrastructure, geography, and trade, making them crucial for developing new technologies and boosting their efficiency. [18]. From 1950 to 1990, seaport development followed a traditional paradigm, with advances in logistics and infrastructure but modest improvements in information and communication technology. Using advanced technology in port operations can benefit both the authority and the community, while also enhancing competitiveness among port authorities. This could lead to the implementation of smart port activities employing cutting-edge technology [9].

From 2000 to today, seaports have upgraded their operating systems to become smart ports with modern technology [19]. According to [20], a smart port integrates terminal operations, warehousing, logistics, yards, and port transportation through a network that provides relevant information for everyday port operations. Ferretti et al. [21] define smart ports as a convergence of IoT, Big Data, automation, and environmentally friendly technologies. This creates a smart infrastructure that connects wired and wireless networks to sensors, allowing data exchange between entities. This would boost logistics and transportation in several industries. Seaport energy consumption is significant due to the high energy demands of port activities. Efficient energy usage poses a difficulty for port officials, as it leads to increased carbon emissions. This will increase the operational costs for such energy [22]. Most port organizations advocate for laws that limit energy use and promote the use of renewable energy resources. This will minimize carbon emissions and energy expenses for port operations. European seaport authorities prioritize air quality as the top environmental priority due to its impact on port operations and maritime activities. To transform ports into smart ports, it is necessary to consider their environmental impact [22]. Molavi et al. [23] added a new domain to the smart port index: safety and security. Seaports' vital infrastructure includes safety and security measures. Recently, there has been a push to improve seaport safety and security through digitization and automation [24]. However, seaport operations have seen a rise in cyber threats [25]. This highlights the need to prioritize safety and security at seaports as one of the four key domains of the smart port index.

2.3 Energy models and energy simulation software

Energy simulation software aims to reduce energy use and associated costs. Identifying sensitive characteristics can assist optimize energy use in properties. Energy simulation tools can assess the influence of new rules and regulations on energy systems, assisting decision-makers in developing business strategies and identifying possibilities based on the evolving system. A simulation of the energy system monitors and analyzes the property's energy use, allowing for fault detection and diagnosis. Simulations can be used to test and deploy new goods for the property, including double glazing, smart cooling systems, LED lighting, and fans. It can also aid in estimating capital costs and energy consumption [26]. Modernization has used various energy sources to enhance human capabilities and innovation. Energy is essential to modern existence. An energy system includes

both the supply sector and the technology used to provide energy services. Energy systems rely on transformed primary energy resources, such as electricity, which is then used by end users for cooling, heating, and lighting [27]. Energy models simplify real-world systems, making testing and experiments more feasible and cost-effective. Computational modeling can have a positive impact by relying on logical assumptions. It can simultaneously interrelate multiple factors. Energy models are classified into six types: optimization models, decentralized energy models, supply/demand-driven models, energy and environmental planning models, resource energy planning models, and neural network-based models [28]. The decentralization and optimization energy models are among the most common. The decentralized energy model aims to meet energy demand using various resources. The goal is to reduce annual energy costs and carbon emissions while increasing overall system energy efficiency. The decentralized energy model can anticipate total energy supply/demand and system costs prior to execution. This assessment will evaluate the impact of social and technological parameters on real-life situations. As a result, the complete system will be implemented and understood well. An optimization model aims to minimize or maximize a function's value under constraints [29].

The Cambridge Dictionary defines simulation as "a situation or event that appears real but is not real, especially used to help people deal with such situations or events" [30]. In the words of Schriber, "Simulation involves the modeling of a process or system in such a way that the model mimics the response of the actual systems to events that take place over time" [31]. Energy simulation tools calculate energy consumption per unit based on various characteristics. This enables simulation users to understand the behavior of the energy system in the unit. Energy simulation software aims to reduce energy consumption, thereby lowering costs. Optimizing energy use in properties involves identifying sensitive characteristics that impact overall energy use [32]. Energy simulation programs examine the influence of new rules and regulations on energy systems, assisting decision-makers in developing business strategies and identifying opportunities. Simulating an energy system allows for monitoring and analysis of actual energy use in a property. This allows for fault detection and diagnosis. Simulations can test and install new items for the property, like new windows, smart cooling systems, LED lighting, and fans. This can help predict capital costs and energy consumption [26]. There are two types of computer-based simulation: discrete event simulation (DES) and continuous event simulation (CES). The DES technique uses discrete steps to solve issues with changing variables throughout time. Meanwhile, CES can be employed for systems with constantly changing variables. DES is a more practical method for analyzing the dynamics of manufacturing systems. CES is ideal for analyzing energy flow in production processes, as loads vary over time [33].

2.4 Gap identification

These are the main gaps identified for this project:

- There is an enormous rise in carbon emissions in the fishing industry
- There is a rise in energy costs and a pressure to address climate change
- Previous research focused on optimizing energy, increasing the efficiency of port machinery/equipment, and minimizing the environmental impact of ships

Seaports serve as the primary hubs for the import and export of products and materials between nations via ships. In business transactions and the acquisition of consumer goods like food, medication, and other items, they are crucial. Many of our contemporary economies are thought to be built upon the crucial and key foundations of maritime transport operations. However, rising carbon emissions pose a serious threat to the world, which is why Western economies have started taking steps to cut carbon emissions through a variety of international accords and alliances. The maritime industry has the potential to significantly reduce carbon emissions, which will have a positive impact on seaports' competitiveness, performance, and quality while also raising the sustainability of the ecosystem in which they operate [49].

The fishing business is facing several significant challenges, one of which is the ongoing increase in energy costs. Given the accelerating work speed and the sharp rise in demand for fish products, this presents a serious problem. This occurs at a time when there is growing international pressure and motivation to address climate change and cut carbon emissions across a wide range of industries. Though several studies suggested specific actions that help somewhat reduce energy consumption and carbon emissions, very few research studies were discovered to highlight the best way to use power energy in ports and fishing buildings. Nevertheless, no research has been done on the potential for creating a comprehensive energy analysis and management system that can be expanded from a site to a community level in order to provide the latter with sustainable economic and environmental advantages. Furthermore, six themes have been identified from the analysis of prior research studies that contribute to the decarbonization of seaports and the fishing industry. These themes include lowering carbon emissions, implementing renewable energy resources, optimizing energy costs, implementing smart control strategies, creating a regulatory framework for greening seaports, and implementing best practice guidelines for greening seaports and the fishery industry. Furthermore, as the ensuing subsections detail, decarbonizing seaports can have a significant impact on port activities [50].

The majority of earlier research has concentrated on creating plans for optimizing the energy mix, increasing the energy efficiency of the machinery and equipment used in ports, and minimizing the environmental effect of ships by lowering their speed and planning when to arrive and depart from seaports [51]. These steps could make a big difference in lowering the carbon footprint of contemporary seaports. Given the complete life cycle of seaports, from design to recycling, lifecycle assessment (LCA) aids in quantifying environmental pressures, trade-offs, and areas for improvement. The highest level of complexity in LCA is found in seaports, which makes it impossible to utilize linear or static methodologies; instead, scalable approaches that take non-linear and dynamic factors into account must be used. Because seaport operations take place over longer time periods than other industries, they must contend with highly different operational and environmental circumstances. For this reason, understanding the ensuing resource consumption and pollution emissions from port activities requires careful consideration of the time dimension in port activity modeling [52].

3 Methodology

The methodology for this project aims to use modeling, simulation, and optimization tools in MATLAB to create an energy management system that can cut costs and emissions for the fish industry while also helping the decarbonization of seaports.

3.1 Creating a simulation using a smart grid

The methodology for this project is developed in the context of the Smart Cluster Energy System for the Fish Processing Industry (piSCES). The piSCES project seeks to lower the prices and the carbon footprint of the fish processing sector by building and testing a new "smart grid" electrical network for fish companies in Milford Haven port.

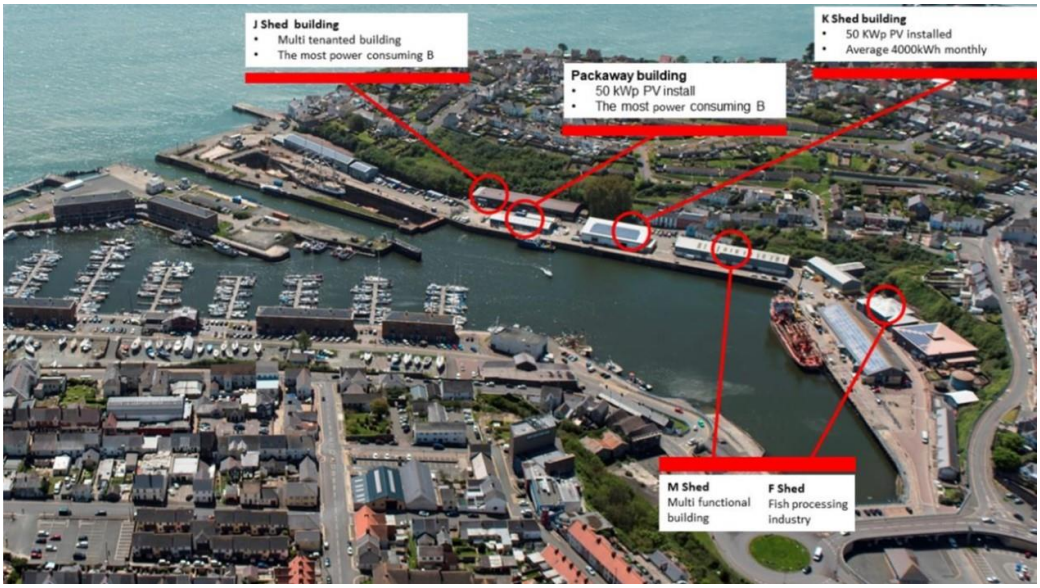

Figure 3. Fish processing industries in Milford Haven port with the solar farm [69]

The smart cluster energy system for the fish processing industry will gradually reduce the cost and carbon footprint of energy networks in the fish processing sector by utilizing smart grid technologies. This will be performed by evaluating their energy networks' usage profiles and optimizing them against the wholesale energy market as well as any potential onsite generation. This study's methodology is generic and applies to the Milford Haven port in South Wales. This port is regarded as the largest energy port in the United Kingdom, as well as the largest oil and gas handler, capable of meeting approximately 30% of the UK's gas demand. One of the port's functions is to ensure the safe transit of vessels along the Milford Haven Waterway, a deep-water site on the UK's western coast. Furthermore, the port has a strategic strategy to diversify its operations, including transportation, energy, renewables, engineering, leisure and tourism, fishing, food processing, and aquaculture. As illustrated in Figure 3, the port contains five primary facilities that generate energy from local PVs connected to the national grid: Packaway, K Shed, M Shed, F Shed, and J Shed. The primary building on the site, Packaway Building, houses energy-

consuming appliances such as ice machines, box washing machines, lighting, and smart meters. The Packaway building contains 50 kW PV panels capable of producing 275 W each panel, powered by two DC-AC converters. Clients use K Shed, a warehouse with a cool room, to store their fish. The cool room is largely powered by 50 kW solar photovoltaic panels. M Shed features an indoor lighting system and various appliances [48].

The smart microgrid incorporated has a load pattern, grid model, energy storage system, renewable energy generation, and control strategies. The integration of simulation components identified critical parameters for deploying a smart microgrid in this application, including battery storage system capacity, energy consumption, and national grid power use. To optimize surplus power distribution in the fish processing industry, a price-based control strategy was developed, taking into account the cost of buying and selling power. This allows energy to be shared within the community rather than sold to the grid. This strategy aims to create a smart energy community at fisheries ports.

Modeling and simulating energy systems can save time and money compared to conducting real-life investigations into energy usage. New energy modeling methods can detect places with high energy use at a certain moment. Additionally, it can accurately model the geometry of real-life buildings. It will also create a comparable environment to the target building. Energy simulation tools provide a sophisticated platform for proposing energy-saving alternatives without affecting the overall system. Changing input variables can assist in studying various scenarios and their impact on the system. This will identify prospective areas for energy savings, cost reduction, and reduced carbon emissions. Energy simulation can assess the feasibility of meeting local energy demands using renewable sources including solar, wind, and tidal power. This will help develop future energy-efficient scenarios and investigate the feasibility of using renewable energy sources to meet local energy demands.

Creating a microgrid platform necessitates various components, including energy generation, demand, storage, energy management, and grid systems. Each system must be developed and tested independently. After constructing a model for each component, the next step is to merge them together into a virtual microgrid. Graphs are then generated to demonstrate the operation of energy consumption. This determines how much power the system receives from the grid and creates during a 24-hour period. It will also be useful in determining the capabilities of specific energy storage devices. The proposed energy control technique increases energy flow and offers information on energy management over a 24-hour period. A finite grid model was used to optimize power flow and prevent blackouts during operations. A finite grid can accommodate unbounded power demand and accept surplus power from the system. After developing a micro-grid model, it's crucial to test it in various circumstances to determine the optimal energy storage for a standalone system. The system must be tested under various settings, including seasons, maximum load, and minimal power production. This will assist in creating a strong model that can be expanded under many scenarios. To improve the reliability of the micro-grid model, it must be run on a powerful computer with ample RAM and clever specifications. This results in a model that is nearly identical to the one to be created on the ground [48].

3.2 Methodology stages

There are 3 simulations run for this project using MATLAB, as described in the methodology section, i) individual building energy optimization ii) cluster energy optimization, and iii) P2P energy optimization. The program utilizes, computes, and generates a large number of variables, with the major goal being to find those that have a substantial impact on final expenses. It initializes the simulation with buildings, appliances, solar panels, and batteries. The main loop iterates over weeks and days, generating weather data to estimate solar production and cost for each building. Energy scheduling involves predicting the demand, utilizing solar energy, charging/discharging batteries, and buying/selling from the grid. Finally, it plots the weekly energy data for each site and calculates the overall costs. Given that the software is meant to regulate energy usage, factors such as solar production and demand have a significant impact on the results. As a result, the algorithm is tested in both typical and extreme life scenarios, such as doubling the demand, to determine its responsiveness. It is investigated which scenario makes the most use of doubled solar production. Furthermore, various solutions are considered for optimizing energy use in response to rapid surges in demand or abrupt drops in energy production. This covers cases in which big changes occur unexpectedly, resulting in crisis situations. In summary, the goal is to create strong solutions that can effectively meet unexpected obstacles and provide appropriate energy management under changing situations.

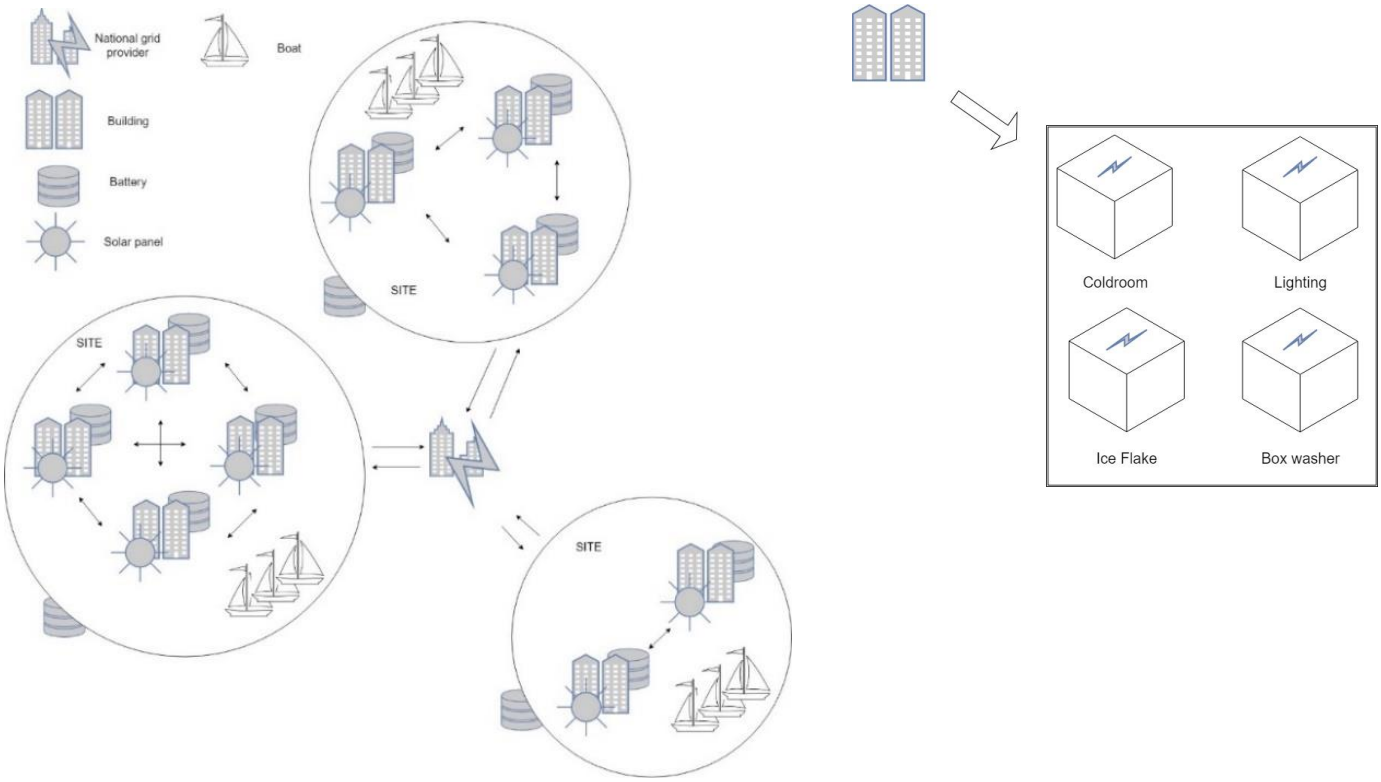

Figure 4: Diagram of the cluster optimization and individual building optimization

Figure 4 shows two levels of consumption. As a cluster, the buildings must share energy in order to reduce trade with the national grid supplier and lower the site's cost. In this situation, buildings can pool their batteries to provide global storage while avoiding the limitations of a building's battery capacity. The fluctuation of market electricity prices, as well as the demand for each building, is easily predictable. As a result, once the optimization is complete, the site needs to fairly divide the cost of each building. Figure 5 depicts buildings with their own bills, and they strive to lower their own expenses. A building will not sell its energy to another building in the same cluster unless it knows it will need it later. These two objectives can be incompatible, which is why many scenarios are investigated to assist the sites and buildings in achieving their goals.

3.3 P2P optimization

After analyzing the micro-grid system and determining its potential for local energy sufficiency at the port level, decision-making mechanisms are optimized inside the network. This improves system efficiency and maximizes the use of local solar energy. This project created and implemented a smart decision-making system using control-based pricing and peer-to-peer control mechanisms. Both smart choice methods result in a near-zero carbon fishery port. Control-based pricing takes into account total energy production from PV panels, battery capacity, and/or electric boats to determine whether to buy or sell energy based on power prices. This will profit the port by taking advantage of high-cost electricity sales or by providing clean, locally produced solar energy to the neighborhood. The P2P decision support system distributes surplus power generated by Agent A to Agent B, thereby meeting local power demand with clean energy. P2P energy sharing can be based on pricing or sharing, which offers additional benefits to local clients. This will allow consumers to become prosumers, distributing surplus energy across the community. Microgrids favor locally produced sustainable energy through clever decision-making algorithms, which benefits both producers and customers. This will have a direct impact on energy markets and society by expanding renewable energy production, empowering customers, and improving access to energy markets [48]. Some of the benefits of using P2P systems include cost savings, environmental sustainability, operational efficiency, and data-driven decision-making. With a rising emphasis on sustainability and renewable energy sources, future directions may include more integration of solar, wind, hydro, and other renewable energy technologies into P2P energy optimization modeling frameworks. By integrating renewable energy generation with energy storage systems and demand-side management tactics, fisheries operations can reduce dependency on fossil fuels and minimize environmental effects.

3.4 Cluster energy optimization

A part of this project aims to model and analyze energy use in the Milford Haven port using a smart energy cluster model, including understanding energy demand and supply restrictions. It involves conducting an inventory of consumption and production units, taking into account carbon emissions and the possibility for the port to become self-sufficient in energy operations. The optimization technique is based on a simulation of the smart energy cluster system. The optimization process involves simulating energy consumption to identify optimal appliance schedules. Simulation-based optimization determines the optimal use of appliances across time intervals using certain parameters and values. The suggested smart grid model connects power

generation to building appliances and uses excess energy to charge a battery system. The system prioritizes energy production and uses it to power appliances. When there is no demand from the building, the battery system is charged as a secondary priority. When there is no demand and the battery is fully charged, any excess energy is transmitted to the national grid. When the PV system generates no electricity, the battery system is used to meet demand. If the battery system discharges during power demand from a building, the model will connect to the national grid to prevent interruption.

A smart energy community can help the fisheries port become more sustainable. Introducing smart energy solutions to the fisheries port reduces energy losses and improves efficiency by using locally generated power. According to Howell et al. [53], implementing a smart-energy community model can benefit an industry by reducing carbon emissions and projecting a positive image in the public and industry. The solar energy generation for this project was modeled in MATLAB using PV panel specifications, site location, and weather data. Data on power consumption and generation were utilized to model a microgrid and optimize its operation for industrial processes. The system includes solar panels with inverters, industries (load patterns), batteries, and the grid. A charge and discharge controller is needed to manage the battery system and its many agents. The simulation model calculates the optimal amount of batteries needed to power the Packaway building. These batteries can encourage stakeholders to adopt power systems that reduce carbon emissions in the fisheries industry. Further research is needed to see if fishery ports can produce positive energy and become zero-carbon champions. Developing and implementing a microgrid for fish companies is a significant step towards improving energy consumption and production, leading to more efficient operations. The grid simulation model covers four seasons. The model was simulated using random intervals in the seasons to analyze the system's behavior under various scenarios. Additionally, it allows for determining the optimal battery storage capacity based on daily charge levels.

4 Evaluation

The evaluation section aims to evaluate the performance of the energy optimization system designed for the fish manufacturing site in decreasing energy costs and carbon emissions under various scenarios.

4.1 The case study

This report focuses on optimizing energy utilization for fish industry agents at Milford Haven, by investigating how the agent-based framework might help to establish smart energy clusters in the industrial environment. This includes optimizing energy consumption, forecasting demand, and dynamically changing energy distribution to ensure sustainability and cost-effectiveness. Milford Haven is a major UK port that handles over 30 million tons of cargo annually. The company, based in the Welsh western coast, specializes in marine operations including cargo handling, renewables, freight, passenger ferry services, fishing, commercial property management, leisure, and retail. Milford Haven's fish docks are perfect for vessels moving between the Irish Sea, Bristol Channel, and Celtic Sea. In recent years, the port has developed fish-processing units with cold storage, ice-making equipment, and box washing machines. Milford Haven port's fish-processing business

struggles with high energy usage as a result of seasonal changes. Milford Haven port's environmental report shows that they consumed approximately 1600 MWh of electricity in 2012 and produced approximately 790 tonnes of carbon emissions. The port authority is installing PV panels to satisfy rising electricity demand and reduce carbon emissions, aligning with the UK's energy plan for 2050. The port authorities deployed 2500 panels across 25 buildings on its premises. Furthermore, the port authorities have established a solar farm at Liddeston Ridge with about 20000 panels and a power capacity of 5 MW. Energy sectors can cut carbon emissions by using local power generation to fulfill demand and avoid relying on the national grid [56]. This project has investigated energy consumption optimization for several fish industry agents at the Milford Haven port in Cardiff, Wales. In recent years, the project has acquired a variety of data on energy production, consumption, and production costs. That is why all of the data used in the model originates from real-time energy monitoring.

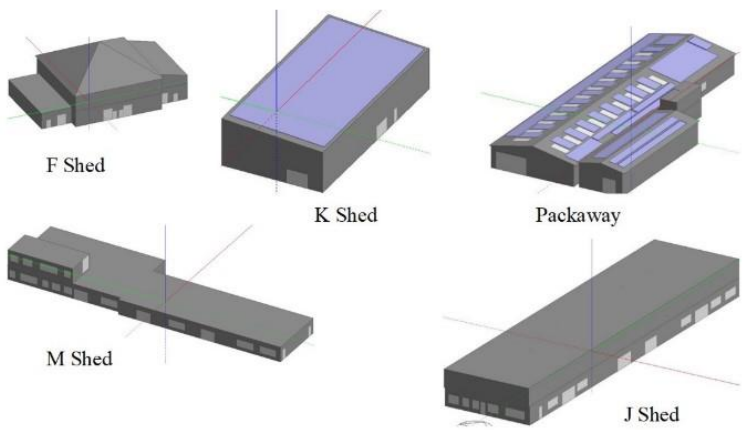

Figure 5. Simulation models of the buildings in the port of Milford Haven [40]

The optimizations are performed based on a variety of characteristics, including the number of sites in the model and the number of buildings within each site. The buildings the port is made out of are shown in Figure 5. As a result, the program makes every effort to accurately depict reality. It investigates several scenarios, ranging from individual agent optimization to peer-to-peer energy trading. Fish industry locations must not only save costs but also minimize their carbon footprint. As a result, the program has been designed to meet the requirements of each situation. The global energy schedule focuses on peer-to-peer exchanges, whereas the daily calendar focuses on energy management. By combining these technologies, we can reduce energy waste and advise buildings on when to utilize their equipment.

4.2 MATLAB simulation and mathematics

The programming for this project was made in MATLAB. Because it supports object-oriented programming and contains mathematical tools built in, including an optimization function, which is the foundation of our project, MATLAB is effective for this project. fmincon is a MATLAB function that is used to minimize a function of many variables under constraints. It is strictly a single objective function. The function accepts numerous arguments, including the objective function, the starting guess for the solution, and the constraints on the design variables. Constraints might take the shape of linear or nonlinear equality or inequality. The function returns both the optimal solution and the value of the objective function at that solution.

As previously said, the program's goal is to schedule a fish industry site's energy usage. The first thing needed to do is to model the simulation. The site is open from 0 am to 6 am, 6 am to 12 am, 0 pm to 6 pm, and 6 pm to 12 pm during the day. There are four appliances in each of these buildings: a box washer, lights, an ice flake producer, and a cold chamber. The fishermen make use of the appliances every day. They visit the facility quite early in the morning before putting their boat out to sea. When the day is cloudy, they only stay for a short while, turning on the fluorescent light tubes to gather boxes that they fill with artificial ice flakes. When they return later in the day, they keep the day's catch in the cold room, a refrigerated space that is maintained at a constant -5°C to maintain the fish's freshness. The boxes they transported to sea that day are then cleaned by turning on the box washing machine. Before being sold, the fish is held overnight in the cold room. If the weather permits, the solar panels installed on the structures provide energy throughout the day. As a result, the buildings have the option to generate their own electricity or purchase it from the national grid supplier. Batteries are another way to store their energy.

The key tool for creating that type of schedule is MATLAB's fmincon function. fmincon aims to discover the x vector that minimizes the function f while satisfying the criteria A.x <= b and A.x0 = b0. A, A0, b, and b0 represent the problem's real-life constraints. It is a highly powerful tool because it can handle linear and nonlinear constraints, quadratic programming, and so forth. (Because MATLAB is used as a programming language, if X is a vector of length N, X(1) is the first element, X(N) is the last, and X(1:N) is an array of all the elements from index 1 to index N). A list of all the limitations needs to be created. To accomplish so, the situation must be correctly modeled. To schedule the energy consumption for the week the only information required is solar production and demand. The fmincon function needs to produce X, a 1x(3*L) vector (L is the number of days till the end of the week). The first L column represents the building's usage. The subsequent L columns represent the energy purchased by the building from the grid provider. The final L columns show the energy sold by the building to the grid supplier. The first restriction states that the building consumption, X(1:day) (where "day" refers to the actual day), must equal the demand. The second constraint is that for each day, the sum of consumption and energy purchased must be less than the sum of solar production and energy sold. The final constraint is that the sum of our purchases and solar production, minus the sum of our consumption and sales, must be less than the battery capacity.

$$A = \begin{bmatrix} \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix} & \begin{pmatrix} -1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ -1 & \cdots & -1 \end{pmatrix} & \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix} \\ \begin{pmatrix} -1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ -1 & \cdots & -1 \end{pmatrix} & \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix} & \begin{pmatrix} -1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ -1 & \cdots & -1 \end{pmatrix} \end{bmatrix},$$

$$b = \begin{pmatrix} \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix} * weekProduction \\ \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix} * weekProduction \quad + batteryCAPACITY \end{pmatrix},$$

$$A0 = \left[\begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \right],$$

$$b0 = weekProduction.$$

$$A.X <= b \text{ and } A0.X = b0.$$

Now that all the restrictions are simulated, for a particular day, fmincon needs to minimize the function.

$$\sum_{i=1}^{L} [X(8 - DAY + i) - 0.9 * X(14 - 2 * [DAY - 1] + i)] * dailyPrice(DAY - 1 + i) \ ,$$

Where dailyPrice is an array that describes the market's energy price fluctuations.

Finally, zeros(1,n) is a matrix 1xn filled with zeros, and fmincon is expressed as follows:

```
[X,~,EXITFLAG,~]=fmincon(@(X)      sum((X(8-DAY+1:2*(7-DAY+1))-0.90*X(15-2*(DAY-1):3*(7-DAY+1))).*dailyPrice(DAY:7)), X0,A,b,A0,b0,

zeros(1,3*(7-DAY+1)),[],@(X) nonLinCon(X,DAY))
```

X0 is the function's beginning point. The remaining three reasons are respecting the lower bound, the higher bound, and what is the final one. If this function is run without the final argument, the result will not be as expected. Indeed, the program will acquire energy even if it then sells it. To address this issue, a non-linear constraint is incorporated into the model, namely L non-linear constraints:

```
for i from 1 to 8-DAY
        X(i+7-DAY+1) * X(i+2*(7-DAY+1) = 0
end
```

These limits prevent the buildings from purchasing and selling on the same day. Finally, the exit flag identifies the function's exit circumstance. Its value only has to be verified now and then to ensure that all optimizations are successful.

4.3 Individual building energy optimization

This simulation assumes that the weather is entirely predictable, which determines the quantity of electricity produced by the solar panels. The prediction of peak demand needs to be considered. By taking into account the known nature of weather and, as a result, solar energy production, as well as fluctuations in market energy prices, peak demand forecasting allows for not only energy storage on days with lower prices but also the opportunity to sell it back to the grid provider during price increases. Furthermore, it has been decided that the boats are associated with one site and consumed on a weekly basis. As a result, the consumption of the boats must be spread between the buildings. Furthermore, the Markovian Decision Process [54] has been used to help the computer plan weekly consumption. The demand is considered as an entity making random decisions, and the outcome of the optimization is determined by these decisions. Data collection also helps to determine solar energy production, which may be thought of as a Markovian Decision Process.

Once the weekly weather forecast and estimated demand are determined, the optimization function "fmincon" is used to schedule building consumption. The available energy for the day is then split across all of the appliances again using the "fmincon" function. Building schedule optimization is used to maximize battery usage, whereas appliance schedule optimization, which is performed daily, increases appliance utilization. Lighting is only enabled if the solar panels do not generate enough energy. When there is plenty of sunlight, illumination is unnecessary, signaling that the solar panels are producing enough electricity. At the end of the day, any excess energy is stored in the battery to prevent it from exceeding its capacity. All daily data is then logged, including information on buildings, appliances, and solar panels. Furthermore, given the development of significant data using random numbers, preservation of this data is conducted to ensure consistent figures are used throughout numerous runs, obviating the need to create new random numbers each time.

While this scenario benefits buildings that can manage their energy use, the lack of peer-to-peer (P2P) trading reduces the effectiveness of this cluster optimization. Without peer-to-peer trade, buildings cannot exchange surplus energy with one another, limiting the potential for optimal energy distribution within the cluster. As a result, if a building is unable to generate enough energy to meet the demands of all of its appliances, it must obtain additional energy from the national grid provider while working independently of other buildings in the cluster. As a result, any energy surplus generated by a structure that exceeds the capacity of its battery cannot be efficiently stored and utilized, resulting in waste. The pseudo-code for this simulation looks like:

```
Initialise every site and every building

for week from 1 to 4

Create random numbers to model the week weather

Save the weather for each building in each site

for day from 1 to 7

for building in site

Use the estimation function for the solar production

Save the daily production for each building

Create the daily demand

Use the prediction function for the demand

Use the function that schedule the energy management

for appliances in building

calculate the daily energy consumption

Schedule the daily consumption division.

end

Upload the new battery

Save the data in the logs

Clear day

end

end

Clear week

End

Display the diagrams
```

4.4 Cluster energy optimization

While traditional energy-building long-term optimization is largely concerned with individual building optimization, this scenario shifts the emphasis to cluster-level optimization. Instead of each building working separately, this technique creates a centralized site to manage the cluster's collective energy demand, solar energy production, and energy storage capacity. The centralized site approach allows for a more coordinated and efficient energy management plan. Rather than applying optimization procedures to each building individually, optimization is performed at the cluster level. Every building in the cluster uses the energy provided by the site in accordance with its priority ranking in the energy queue. The priority order is decided by three factors: the time a building has purchased energy, the amount of energy purchased, and a random rank issued during setup. Buildings that have purchased more energy for longer periods of time are given a higher priority in the queue. This prioritizing guarantees that buildings with higher energy requirements receive priority in energy allocation. In circumstances where the site generates more energy than the buildings demand, the excess energy is distributed fairly across all structures in the cluster. If the site decides to sell extra energy back to the grid, the proceeds are split among all buildings. These elements, together with the rank-order priority mechanism, ensure that energy purchases and benefits from energy sales are evenly divided among all participating buildings. Implementing this cluster-level optimization strategy improves the efficiency and effectiveness of energy management within the cluster. Centralized coordination of energy resources, as well as equal distribution of energy purchases and rewards, help to create a more balanced and sustainable energy environment for all buildings involved.

The key difference between this scenario and the first is that all optimization functions are applied to the global building rather than each building individually. In theory, this scenario has several advantages over the first. The centralized site can compensate for the restricted capacity of individual building batteries. Furthermore, the site uses expected demand to estimate battery requirements. However, in the event of an error, the site may be forced to sell energy at a disadvantageous moment. Despite the absence of actual peer-to-peer (P2P) trading, charge redistribution helps to reduce overall cluster costs. Nonetheless, this scenario is based on an idealized situation. In practice, buildings have little incentive to exchange energy if it increases their costs. The pseudo-code for this simulation looks like this:

```
Initialise every site and every building

Initialise a global building

for week from 1 to 4

Create random numbers to model the week weather

Save the weather for each building in each site

for day from 1 to 7

for building in site

Use the estimation function for the solar production

Define the global production as the sum of every

 building production

Create the daily demand

Define the global demand as the

 sum of every building demand

Use the prediction function for the demand

of the global building

Use the function that schedule the energy management

of the global building for appliances in building

The building solar energy
available is the

global solar energy
remainingcalculate the daily

energy consumption

Schedule the daily consumption
division.

The global energy available is
subtracted by

 the building consumption

end

Divide the purchases and the
sales with all the

buildings

Upload the new global battery

Save the data in the logs

Clear day

end

end

Clear week

End
```

4.5 Peer-to-peer energy optimization

The main aim of these simulations is not only to reduce the cost of the cluster but also to reduce the carbon footprint of the site. Leveraging peer-to-peer (P2P) trading allows us to reduce our dependency on transactions with the national grid provider, resulting in lower carbon emissions. In this case, intra-cluster trading is prioritized above exterior trading. Once again, our data center is used to estimate daily demand and weekly solar energy production, which is then optimized

with our function. However, before scheduling appliance consumption, buildings try to engage in mutual trading. Building A can only swap energy with Building B if the latter has already scheduled all of its appliances and acquired surplus energy at a price equal to or less than the current day's price for usage the next day.

This condition promotes intra-cluster exchanges, which reduces external transactions and carbon emissions. Because this scenario prioritizes intra-cluster exchanges, it may prevent buildings from fully charging their batteries. This is a problem for structures attempting to forecast periods of surplus energy for sale. Furthermore, an issue develops when multiple buildings engage in internal energy transactions, essentially buying and selling energy within the cluster on the same day. To remedy this, limits were put in place to prevent similar incidents from occurring. Another disadvantage of this scenario is its reliance on solar energy generation as the primary input for optimization. As a result, opportunities for electricity exchange between buildings are limited. If a building can adequately power all of its appliances, other buildings are unlikely to have excess energy available for trade. The pseudo-code for this scenario is very similar to the initial one, with the inclusion of trade optimization with the national grid supplier after weekly energy consumption scheduling, therefore the following pseudo-code is added:

```
for each building

for each building B that already scheduled its consumption

if B has stored more energy than it has consumed

if building has to buy more energy than B stored

building can buy from B

end
```

5 Results

After running the program, MATLAB returns the cost, CO2 emissions, and a graph for the week’s schedule. The dashed line represents the building's energy demand, which is the projected energy usage for the week. Notably, the diagram includes solar energy generation because all future consumption and optimization functions rely on the availability of solar energy. As a result, the sales and purchases depicted are direct outcomes of the building schedule optimization, and neither can deviate from zero on any given day. Furthermore, the battery is an important component of energy management, so it is represented alongside the other sections on the graph. All the analysis was done by comparing the modifications to the same configuration. It was assumed that a typical site consists of five buildings, each with a battery capacity of 130 kW, and just one boat. As previously said, the data is generated randomly using the Markovian Decision Process based on the current consumption of the port. Based on the Customer Relationship Management [2], plenty of data was gathered about the energy consumption of a building and the consumption of each of its appliances in order to predict how the demand will evolve through the week. The demand can be seen as an agent who makes random decisions and where the result of the optimization depends on these decisions. The gathering of data also allows to determine solar energy production, which can as well be seen as a Markovian Decision Process. For each of the scenarios, to do a multi-run

analysis it should run first with growth = 1, and solar growth = 1, because the number multiplying the demand and the solar production depends on the first run made. All three scenarios have the same inputs and similar outputs. The objectives are to reduce the cost and the CO2 emissions. The demands were chosen to closely match the actual energy consumption patterns found in the fishing industry and seaports. Realistic demand patterns are critical for assessing the efficiency of optimization algorithms in real-world settings. The cost of energy is not constant but rather varies over time. The buildings anticipate high demand and are able to limit exchanges with the national grid supplier as prices rise. The optimization functions seek to reduce energy expenses by timing appliance usage and trading based on variable energy prices.

5.1 Scenario 1

The first scenario describes the help of a simulation to optimize the energy for each building. In this case, the buildings on the site want to reduce their own cost, by using their own batteries for energy storage and individual collaboration with the grid.

5.1.1 Double energy demand

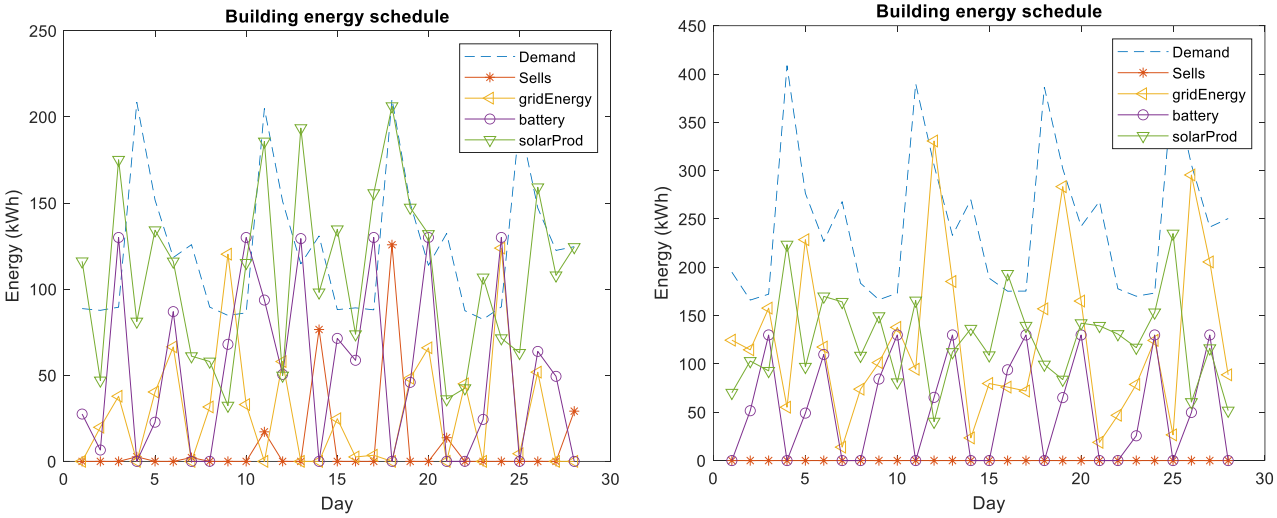

Figure 6: Energy demand in normal conditions (left), Energy demand when doubled (right)

For the first scenario as the site contains five buildings, five different graphs appear. However, in this article only one of them is analyzed, corresponding to the main building, The Packaway building. The first graph shows the initial energy demand and solar production with a battery capacity of 130 kW for each of the buildings. The second figure shows the outcomes of doubling the energy demand.

Figure 6 shows the energy demand in normal conditions, predicted randomly [2] [54], and the results when the demand is doubled. The results of the first run show in MATLAB that the initial average cost of each of the buildings came to £45 and the cost altogether came up to £223. This simulation is purposely run for the reason that each building on the site wants to reduce its own cost. Regardless of the number of buildings per site or the number of boats, doubling the demand

raises the cost by an average of £500 per building. The cost of each building will rise as the buildings seek to reduce their own energy cost. As a result, the cluster will cost more as the number of buildings increases. From the results, battery capacity has no impact on building costs. However, doubling the battery capacity when the demand is double that of a regular condition saves the cost of a structure by £50. The first graph shows peak energy demand on day 4 at 210 kWh and the lowest demand on day 23 at 80 kWh. It can be seen that when the demand doubles, solar production cannot meet the needs of the buildings. Therefore, the cluster will frequently need to purchase electricity from the grid supplier in order to power its gadgets, if the solar production is insufficient. This results in an increase in the cost, as shown above. The peak demand stays at 408 kWh on day 4 and it's lowest at 166 on days 2 and 9. Nonetheless, the buildings anticipate strong demand and are able to limit their transactions with the national grid supplier when prices rise. Furthermore, the environment responds fast to a significant shift in the demand. When the day's demand deviates significantly from what was projected, the algorithm expects that regular behavior will not be followed and that the gap must be filled between the previous predictions and reality. Thus, with the cluster's limits, the program is able to anticipate new spikes in demand and fill the batteries when the price of electricity is low, lowering the cluster's cost as much as feasible. The energy demand throughout the scenarios is shown to be in a constant pattern throughout the days. It starts at a low point, reaches its maximum, and then reduces over a period of three days. This happens because the simulation considers perfect weather and simplified models for consumers. In a real-life situation, the energy demand would keep increasing in an irregular pattern.

5.1.2 Double Solar Production

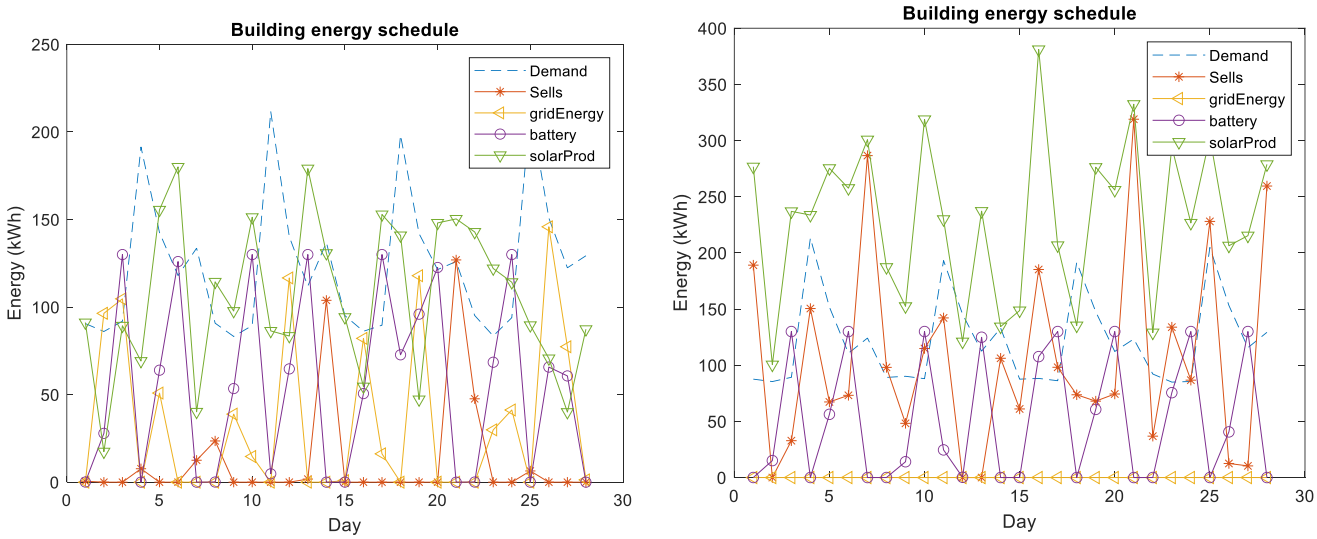

Figure 7: Solar production in normal conditions (left) Double Solar production (right)

Figure 7 compares the data between normal solar production and doubled production. The results of the first run show in MATLAB that the initial average cost of each of the buildings came to £76 and the cost altogether came up to £379. The top solar production on the graph is at 180kWh on day 6, and the lowest is at 17 on day 2. Whereas the cost increases when demand doubles, the

cluster can make a lot of money when solar production doubles, reducing their reliance on the national grid. When the solar production doubles the peak goes up to 381 kWh on day 16, and the minimum at 101 kWh on day 2. In this case, a building can generate up to £516 each month on average and about £2582 as a cluster. However, even if the cluster can become richer in this instance, as long as there are buildings on the site, capacity restricts the amount of money the cluster can earn. As a result, if the battery was considerably larger, the buildings would be able to preserve the energy and sell it when the price increases. Indeed, most of the time, a building will have to sell practically all of the excess energy it generates since its battery cannot hold it. However, by selling extra energy back to the grid or engaging in energy trading markets, buildings can profit on favorable pricing conditions and perhaps offset their energy costs further.

5.1.3 Double battery capacity

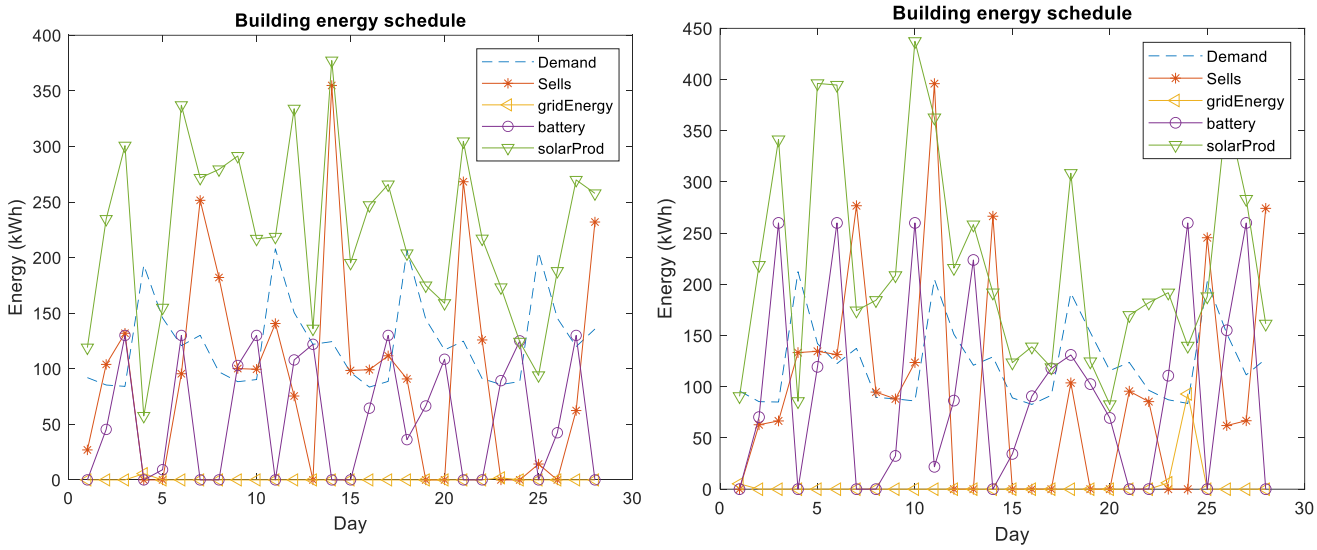

Figure 8: Battery capacity of 130kW (left) Battery of 260 kW (right)

In this scenario, Figure 8 shows that when the battery is equal to 130 kW, solar production peaks on day 14, hitting 380 kWh. On day 4, however, solar production reaches its lowest point, falling to a meager 57 kWh. It can be seen that the building sells the most amount of its energy on the day when there is the most energy production. On this day, the building generates far more energy than it consumes. Thus, it must sell a lot of energy in order to avoid exceeding the battery capacity limit. Then it sells less energy on the more valuable day. On the contrary, where the battery capacity is 260 kW, on the peak solar production day, the building does not sell most of the energy it produces. The battery conserves this energy in order to earn more money the next day. Larger battery capacity increases buildings' resilience to variations in energy supply and demand. Buildings can better handle intermittent Solar Production energy and unanticipated changes in energy usage patterns, ensuring a more consistent and stable energy supply within the cluster. Furthermore, increased battery capacity allows buildings to adjust their energy consumption in response to the national grid or market prices. Buildings can benefit from incentives for lowering energy consumption during peak hours or maintaining grid stability during times of high demand.

5.4 Scenario 2

The second scenario, cluster energy optimization, prioritizes intra-cluster energy trading over grid trading. In this situation, optimization functions are applied to clusters rather than individual buildings. The centralized site compensates for the restricted battery capacity of separate buildings, and it uses predicted demand to estimate battery usage and scheduling.

5.4.1 Double demand

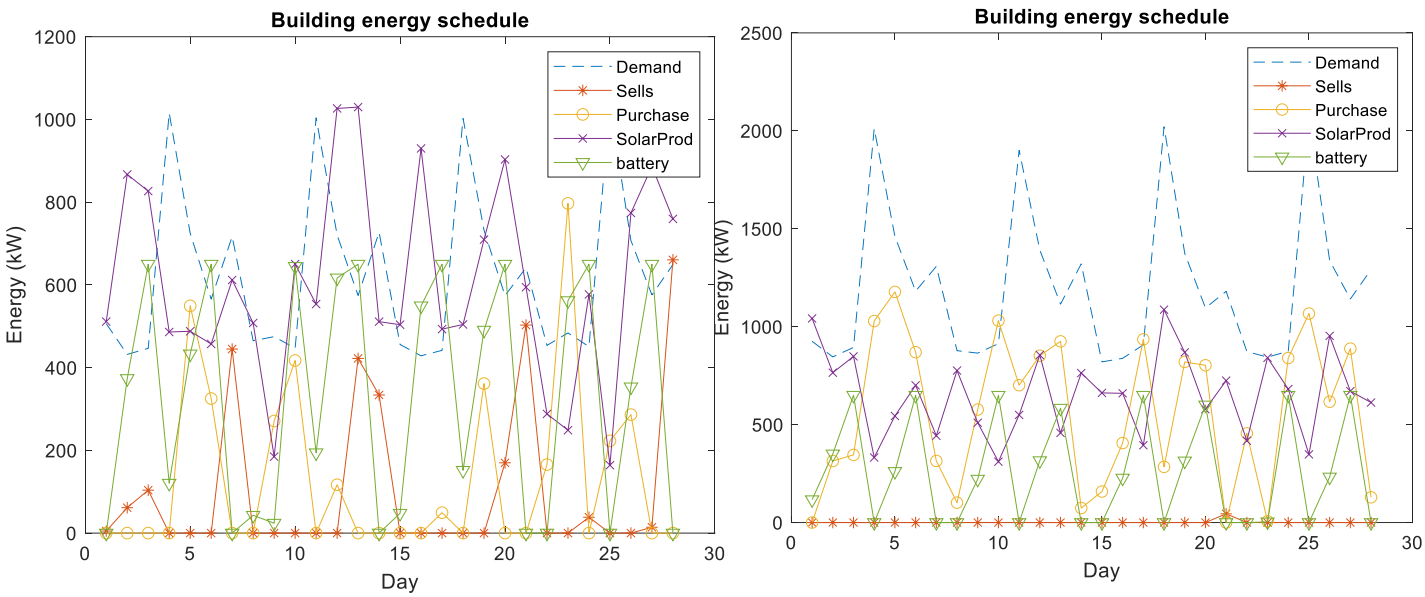

Figure 9: Energy demand in normal conditions (left), Energy demand when doubled (right)

In the second scenario, the first outcomes are unexpected at first. The results of this scenario are so similar to the results from the first scenario that they can be considered identical. The cost of the cluster came to £2582 for scenario 1 and £2658 for scenario 2. This can be explained by the fact that the second scenario's strength is predicated on its ability to assemble the batteries, and therefore to deal with one building's limited battery capacity. However, as shown in Figure 9, when the demand is doubled, it exceeds the capacity of a building's battery. As a result, this scenario becomes equally valuable as the first. From the second graph, it is visible that the energy production is insufficient compared to the energy demand. The simulation would need to buy more energy from the national grid in order to accommodate the increased demand. Similar to the first scenario the battery storage capacity would need to be bigger to meet the increased demand and stabilize energy supply. Compared to the first graph, the cluster does not sell any of its energy when the demand is doubled, as the solar production is low. The energy production needs to be doubled to meet the demand and cut ties with the grid.

5.2.2 Double Solar Production

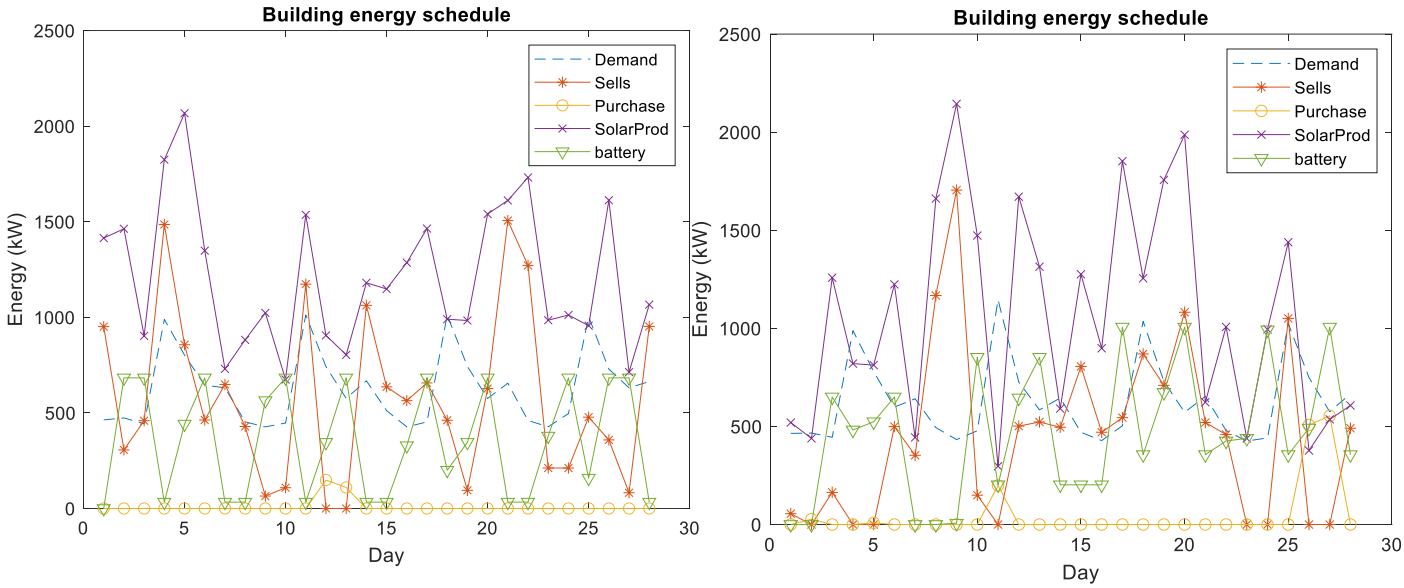

Figure 10: Solar production in normal conditions (left) Double Solar production (right)

As shown in Figure 10, the battery limits the cluster's ability to become as rich as it could with a larger battery capacity. This effect is considerably more visible in this instance since the battery is the key advantage in order to cut costs. The highest energy production occurs on day 9 at 2200 kWh and also sells the most that day. In the next few days, the cluster sells less energy in order to not overflow the capacity of the battery. With solar production doubled the cluster becomes less dependent on purchasing energy from the grid and lowering energy costs. However, the battery capacity needs to be increased during periods of peak demand or low solar availability, to enhance the overall energy resilience. During peak solar production there is excess energy generated that needs to be stored in batteries for cost efficiency or sold back to the grid. Comparing the results of the initial solar production in this scenario with the first scenario there is a significant change in the purchase from the national grid. Looking at this scenario there is almost no purchase with the grip throughout the 28-day period, except for when the production is at its lowest. This is proof that cluster optimization is more efficient than when the buildings try to decrease their cost individually. It takes a centralized approach coordinating all buildings, enabling equal distribution of energy/cost benefits.

5.2.3 Double battery capacity

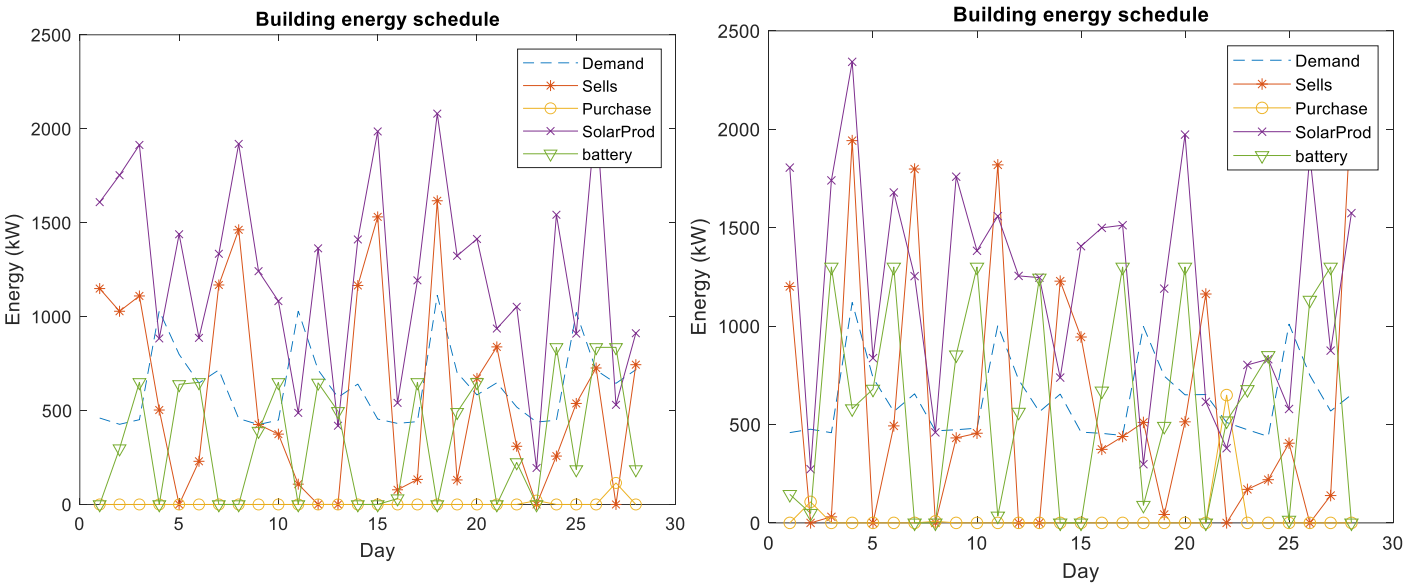

Figure 11: Battery capacity of 130kW (left) Battery of 260 kW (right)

With double battery capacity, the cluster stores more surplus energy generated during periods of high solar production. As shown in Figure 11, when the energy demand stays the same, but the solar production and battery capacity double the cluster sells more energy. This benefits the cluster in terms of cost savings. By storing excess energy and reducing reliance on grid power during peak hours, the cluster could realize significant long-term cost savings through a more efficient and cost-effective energy management strategy. Furthermore doubling the battery capacity gives the opportunity for the surplus energy generated during peak solar hours to be stored in the larger batteries and discharged during periods of high demand or low solar production. This capability optimizes energy usage and reduces peak demand charges. Even in this case the connections with the grid are almost completely cut off. The building sells more energy saving a lot in terms of cost.

5.3 Scenario 3

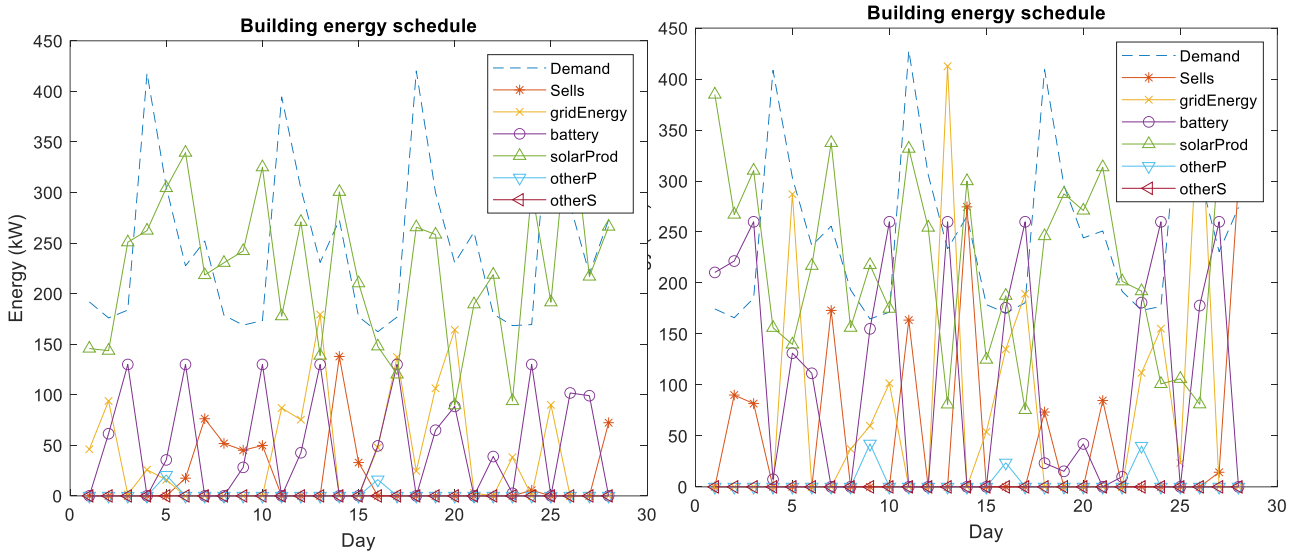

Figure 12: Battery capacity of 130kW (left) Battery of 260 kW (right)

The third scenario, peer-to-peer (P2P) energy optimization, prioritizes intra-cluster trade above external grid trading. In this scenario, buildings within the cluster try to engage in reciprocal trading. Building A can only swap energy with Building B if B has already scheduled its appliances and has extra energy at a price equal to or lower than the current day's pricing. This encourages intra-cluster exchanges, reducing external transactions and carbon emissions. This scenario focuses on minimizing trades with the national grid provider, hence reducing CO2 emissions. Although it does not reduce the cluster cost, and in certain cases, the cluster cost is slightly greater than scenario 1, the CO2 emissions are much lower than in other scenarios. This scenario gains strength by facilitating trades within a cluster. However, as seen in Figure 12, if solar production is twice as high as usual, buildings will struggle to control their energy use. To avoid exceeding the battery limit, each building will be required to sell any excess energy it generates every day. As a result, each building will be unable to purchase energy from the other buildings. Similarly, if demand doubles, each building will have to buy from the national grid provider and will be unable to sell their electricity to other buildings. This restriction on intra-cluster trading could diminish the advantages of this scenario during periods of high demand. Nevertheless, by doubling the battery capacity in this scenario it can be seen that three V-pikes are representing the moments this building decided to buy energy from other buildings instead of the national grid provider. Increasing the battery capacity results in higher energy trading within the cluster. Thereby, this reduces the trades with the outside of the cluster, hence that reduces the CO2 emissions and the costs. However, the graph shows a high reliance on the grid when solar production is insufficient, making this scenario less efficient than the second one. P2P in theory, aims to promote intra-cluster energy exchanges for cost savings, however, the simulation in the third scenario prioritizes the decrease of CO2 emissions rather than the cost savings.

6 Conclusion

6.1 Summary

Developing energy systems in seaports and fishing sectors is a promising prospect. This can be achieved by using modeling and simulations to understand the characteristics of energy usage. Implementing a smart grid method can improve energy management and performance. This improves energy efficiency and lowers long-term costs. Smart energy systems in seaports can transform fisheries and industries by increasing awareness of energy systems, increasing efficiency, automating processes, and utilizing available resources. The development of port energy systems will encourage decision-makers to convert seaports and fishing ports into more efficient and cost-effective facilities. This will help build policies to reduce the use of fossil fuels. Seaports can attract investment from energy firms and investors by participating in tenders to update energy systems with new technology, leading to long-term gains. Seaports are a new area for research and development, as there has been limited research in this field. The development of port energy systems will promote efficient utilization of seaports and fishing ports, creating a clear vision for decision-makers. This initiative will bring together bright minds and researchers to create a new vision for seaports and fishing sectors by implementing digital technologies for port infrastructure and operations.

6.2 Conclusion of results

This study presented a simulation design to optimize energy consumption and reduce fish costs for a fish industry site, Milford Haven in Cardiff, Wales. The findings show that using MAS can help achieve the primary goals of improving energy efficiency, lowering costs, and contributing to the decarbonization of fish industries. The individual building optimization scenario demonstrated cost savings and lower CO2 emissions by optimizing appliance consumption, harnessing solar energy, and controlling battery storage. The cluster optimization strategy resulted in even larger cost savings by combining battery storage and coordinating energy distribution across the cluster. The peer-to-peer (P2P) energy trading scenario increased efficiency by allowing buildings to trade surplus energy and lowering reliance on the national grid, thereby contributing to the decarbonization goal.

The simulation aimed to reduce the cost of fish processing by analyzing individual, cluster, and P2P optimization. However, cluster optimization is challenging to implement in real-life situations, even though enables significant cost changes, unless all the buildings are owned by the same company. This study acknowledges that several assumptions and simplifications were made in the optimizing functions, necessitating careful analysis of the results. Further research is required to address limitations such as the simplified model for consumers, the oversimplified weather conditions, and the seasonal variations in solar energy production estimates.

The cluster optimization technique prioritizes energy consumption for neighboring buildings in microgrids over selling to the national grid. Simulation results based on these constraints show that energy sharing allows prosumers to share surplus energy in smart energy communities, resulting in monetary advantages and effective resource utilization. The study found that using cluster optimization scenarios improved overall agent performance in the smart grid. The price-

based control algorithm utilized an energy storage device to prevent energy losses and maintain energy levels in the system. In the peer-to-peer scenario, the control algorithm demonstrated that surplus energy generated by prosumers can be used to meet local energy demands, before selling to the grid. These strategies can encourage local renewable energy generation in the smart energy community. This study found that using price-based control and peer-to-peer methods can significantly reduce carbon emissions from traditional fossil fuel consumption, contributing to the decarbonization of seaports.

Despite these limitations, the study offers valuable insights for fish industries in managing energy consumption and CO2 emissions. It provides recommendations on when to trade with the National Grid, schedules appliance consumption throughout the day, and returns an optimized schedule to minimize costs for individual buildings and as an entire cluster. Additionally, the algorithm enables the study of the impact of various environmental factors and different cooperation scenarios amongst buildings. The program was designed to facilitate the implementation of modifications, allowing for further refinements and improvements. While the study presents a promising approach to energy optimization in fish industries, there is still a need for further research and refinement to enhance the accuracy and applicability of the simulations. Future research with local authorities, including the Milford Heaven Council, aims to explore these strategies. The optimization technique aimed to minimize equipment deployment and investments while improving energy management at the site and community levels.

6.3 Significance of work

The primary importance of this study is that it demonstrates how advanced modeling and simulation techniques may be used to optimize energy systems in seaports and fishing businesses, resulting in increased energy efficiency, cost savings, and lower carbon emissions. The project looks at the use of multi-agent systems and optimization algorithms to model and simulate energy use, production, and trading at the individual and cluster levels. This enables cluster energy optimization, peer-to-peer energy sharing, and increased usage of renewable energy sources such as solar power. The results demonstrate how these simulation-based approaches can help transform fishing industries into more sustainable and cost-effective operations by identifying potential for energy savings, pollution reductions, and enhanced resource management. The project creates a complete model to simulate complex energy systems, including the dynamic interplay of energy generation, demand, storage, and grid systems. This is consistent with recent research on modeling and optimizing energy systems utilizing agent-based techniques and multi-agent systems.

6.4 Rigor of research

The project's modeling and simulation approach is quite rigorous. The researchers used complex MATLAB simulations to simulate and improve energy systems, taking into account a wide range of real-world parameters. The simulation models were evaluated against real-world data to guarantee that they appropriately represent actual energy trends. The results were presented concisely, demonstrating the advantages of various energy management measures. Overall, the research shows a strong and thorough investigation into optimizing energy systems in fishing industries.

6.5 Possible impact

The simulation models and optimization approaches created in this project have the potential to change the seaport and fishing sectors into more efficient and sustainable operations. The ability to quantify and optimize energy use, production, and trade at the individual building and cluster levels can assist these industries in identifying areas for significant cost savings and emissions reduction. The models' rigorous evaluation against real-world data from the Milford Haven port reveals their ability to properly represent actual energy trends and guide practical implementation. The research looked into the potential for MAS to integrate renewable energy sources into the energy management system. The project addressed a gap in the use of renewable energy sources by modeling them as agents, supporting the development of smart energy communities within seaports.

Overall, the study demonstrates how advanced modeling and simulation can be used to update energy systems in seaports and fishing facilities, resulting in more cost-effective and environmentally friendly maritime operations.

6.6 Future work

The dissertation acknowledges the limits of the models, such as oversimplified weather conditions, and emphasizes the need for additional research to fill these gaps. This transparency and understanding of limits pave the way for future research to improve the accuracy and breadth of simulations and optimization methodologies.

Energy system optimization is not a novel concept but most approaches consider the system as a static entity, focusing just on a single system rather than adopting a comprehensive picture of the dynamic system landscape. Researchers argue for a system-of-systems approach to energy management, utilizing semantic models to provide a comprehensive understanding of energy systems and their socio-technical components. This is crucial for addressing several scenarios, including local energy balance, air-landing, and blackout prevention, in a rapidly changing world with highly distributed energy resource penetration. A semantic framework can address flexibility, scalability, robustness, openness, and practicality. It's crucial to balance the benefits of distributed control and centralized control while addressing their respective downsides. Semantic and AI-based design offers significant benefits for restoring and healing the grid following service disruptions. The grid should be able to restore, reorganize, and repair itself through alternative

topologies without harming the system as a whole, guided by a holistic awareness of the larger energy systems.

The energy system can be reconfigured dynamically to optimize performance, enabling demand-responsive energy management. Common techniques for power system management include optimization-based grid planning and long-term operational control. Increasing stochasticity on supply and demand, along with the connectivity of multiple sectors and markets, complicates power grid operation and necessitates multi-objective, time-series-based optimization in uncertain environments. Increasing data availability and computer capacity allows for short-term optimization and near-real-time operations.

This study not only examines energy behavior over time but also saves time and money on real-world energy usage investigations. New energy modeling technologies pinpoint places with high energy use at a certain period. Additionally, it enables accurate modeling of real-world building shapes and components. Energy simulation tools provide a sophisticated platform that allows to propose numerous energy-saving scenarios without affecting the entire system. Changing input variables can assist study different situations and assess their impact on the entire system. This identifies promising ways to save energy, lower costs, and minimize carbon emissions. However, as a virtual environment, simulations cannot affect real-world energy use. Understanding energy use in buildings can assist in identifying key characteristics affecting the energy system. Energy simulation can assess the feasibility of meeting local energy demand using alternative sources like solar, wind, and tidal.

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Appendix A – Record of project meetings

CARDIFF SCHOOL OF ENGINEERING

NAME..Fiona Topore.....**STUDENT NUMBER**.....21058513.....

SUPERVISOR.....Ioan Petri.....

RESEARCH GROUP ...BRE.....

Date of meeting	Supervisor's assessment of progress	Actions by next meeting	Supervisor signature
01/02/2024	Good	Introduction to the project	
22/02/2024	Good	Talk about the structure of the project	
27/02/2024	Good	Review project progress	
16/03/2024	Good	Review methodology section	
22/04/2024	Good	Project presentation	
25/04/2024	Good	Update on the project	
30/04/2024	Good	Tips on finishing up the project	